

Parameterizing the genetic architecture under stabilizing selection

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Abstract

Across many complex traits, genetic variants with larger effect sizes tend to occur at lower frequencies, which is often interpreted as a signature of stabilizing selection. In statistical genetics, the so-called α -model captures this relationship by assuming that effect size variance is inversely proportional to heterozygosity raised to a power $0 \leq \alpha \leq 1$. Although empirically useful, the α -model is phenomenological rather than mechanistic and lacks a direct population genetic interpretation. In this paper, we derive an alternative to the α -model based on evolutionary theory. Our approach yields a linear mixed model in which frequency dependence emerges naturally as a function of interpretable evolutionary quantities describing mutational variance, selection intensity, and coupling between the focal and selected traits. These quantities enter through two identifiable variance components that can be estimated by restricted maximum likelihood (REML). The resulting framework links a fitness landscape model to standard mixed-model methodology, enabling both inference on evolutionary parameters and downstream prediction by best linear unbiased prediction (BLUP). In forward simulations, the model accurately recovers the focal-trait variance and generally improves genetic prediction relative to conventional α -model baselines.

Keywords: Stabilizing selection; Polygenic trait; Statistical genetics; Restricted maximum likelihood; Linear mixed model;

Introduction

A fitness landscape maps genotypes or phenotypes to reproductive success and provides a geometric framework for reasoning about how selection and mutation shape evolutionary change (Tenaillon 2014; de Visser and Krug 2014). This concept has been used to study adaptation and predict evolutionary trajectories, to characterize epistasis and landscape ruggedness, and to understand how genetic variation is maintained under competing forces of mutation, selection, and drift.

The canonical example of a fitness landscape is Fisher's geometric model (FGM), which supposes that an individual's fitness is given by

$$\text{fitness} \propto \exp\left(-\frac{\|z\|^2}{2V_S}\right) \quad (1)$$

where $z = (z^1, \dots, z^d)$ is a d -dimensional trait vector subject to stabilizing selection and V_S controls selection intensity (Fisher 1930; Tenaillon 2014). Originally proposed by R.A. Fisher almost a century ago, FGM has seen renewed interest over the past decade as a theoretical framework for understanding trait-associated loci in genome-wide association studies (GWAS). By considering a special instance of FGM with isotropically distributed mutational effects, Simons *et al.* (2018) made predictions about the distribution of GWAS-derived effect sizes. Another study used this framework to estimate the contribution of complex traits to fitness in actual GWAS data (Simons *et al.* 2025). Using predictions derived from FGM, Koch *et al.* (2024) further argued that stabilizing selection is a primary mode of selection acting on human traits, and gave estimates as to its strength.

These works help explain repeated empirical observations from large GWAS of a negative correlation between effect size and allele frequency (Zeng *et al.* 2018; Schoech *et al.* 2019; Privé *et al.* 2020). Generally, this correlation is ascribed to purifying selection, where large effect variants are more likely to be removed from the population. To quantify this relationship, a common choice is the so-called α -model, which supposes that the variance of effect size is inversely proportional to heterozygosity raised to a power $0 \leq \alpha \leq 1$. This model first appeared in the setting of estimating genetic relatedness matrices from single nucleotide polymorphism (SNP) markers, and was later extended to explicitly detect negative selection (Zeng *et al.* 2018; Schoech *et al.* 2019).

Although it captures the underlying relationship well, the α -model is largely based on statistical curve fitting. To the best of our knowledge, a population genetic model that predicts this relationship has yet to be developed (Sella and Barton 2019; Koch and Sunyaev 2021). In this work, we fill that gap by proposing a general procedure for combining an evolutionary model with mixed-effects models in statistical genetics. This yields a linear mixed model parameterized by evolutionary quantities that can be estimated with restricted maximum likelihood (REML; Johnson and Thompson 1995). These quantities appear in the variance of random effects encoding the frequency dependence of effect sizes, which is distinct from the α -model. Unlike the α -model, the new formula does not diverge as the sample frequency approaches zero, and allows unbiased estimation of mutational variance. Furthermore, as a mixed-effects model, it can estimate genetic values using the best linear unbiased

1 predictor (BLUP).

2 Theory and model

3 Diffusion approximation for stabilizing selection

4 Assuming that individual fitness follows the FGM (1), the
5 marginal selection coefficient s_{eff} acting on a given locus is

$$s_{\text{eff}}(x_j) = \left(x_j - \frac{1}{2}\right) \frac{\|\alpha_j\|^2}{V_S} \quad (2)$$

6 for a biallelic locus j of frequency x and effect size $\alpha_j =$
7 $(\alpha_j^1, \dots, \alpha_j^d)$ with α_j^k corresponding to z^k (Robertson 1956; Lande
8 1976; Simons *et al.* 2018; Negm and Veller 2026). Hereafter,
9 α_j denotes the vector of effect sizes of the j -th locus for traits
10 $k = 1, \dots, d$, and $\alpha^k = (\alpha_1^k, \dots, \alpha_p^k)$ denotes the vector of effect
11 sizes of the k -th trait across loci $j = 1, \dots, p$.

12 In a panmictic diploid population of size N , the diffusion
13 approximation corresponding to (2) is

$$\begin{aligned} dX_{t,j} &= (s_{\text{eff}}(X_{t,j})X_{t,j}(1 - X_{t,j}) + \mu(1 - 2X_{t,j}))dt \\ &+ \sqrt{\frac{X_{t,j}(1 - X_{t,j})}{2N}}dB_t \\ &= \left[X_{t,j}(1 - X_{t,j}) \left(X_{t,j} - \frac{1}{2} \right) \frac{\|\alpha_j\|^2}{V_S} + \mu(1 - 2X_{t,j}) \right] dt \\ &+ \sqrt{\frac{X_{t,j}(1 - X_{t,j})}{2N}}dB_t \end{aligned} \quad (3)$$

14 where B_t is Brownian motion. Rescaling time in units of $2N$
15 generations, with $V_S = 2N \cdot W_S$ and $\theta = 4N\mu$, we obtain

$$\begin{aligned} dX_{t,j} &= \left[X_{t,j}(1 - X_{t,j}) \left(X_{t,j} - \frac{1}{2} \right) \frac{\|\alpha_j\|^2}{W_S} + \frac{1}{2}\theta(1 - 2X_{t,j}) \right] dt \\ &+ \sqrt{X_{t,j}(1 - X_{t,j})}dB_t. \end{aligned} \quad (4)$$

16 The forward Kolmogorov equation corresponding to (4) is

$$\begin{aligned} \frac{\partial p(x_j, t; \|\alpha_j\|)}{\partial t} &= \\ &- \frac{\partial}{\partial x_j} \left[\left(x_j(1 - x_j) \left(x_j - \frac{1}{2} \right) \frac{\|\alpha_j\|^2}{W_S} + \frac{1}{2}\theta(1 - 2x_j) \right) p(x_j, t; \|\alpha_j\|) \right] \\ &+ \frac{1}{2} \frac{\partial^2}{\partial x_j^2} \left[x_j(1 - x_j) p(x_j, t; \|\alpha_j\|) \right]. \end{aligned} \quad (5)$$

17 Equations (4) and (5) are equivalent representations of the same
18 stochastic process $X_{t,j}$.

19 At stationarity, $\partial p / \partial t = 0$ and we obtain (Ewens 2004)

$$\begin{aligned} p_{\infty}(x_j; \|\alpha_j\|) &= \\ &= C(\|\alpha_j\|)^{-1} \cdot x_j^{\theta-1} (1 - x_j)^{\theta-1} \exp \left[-\frac{\|\alpha_j\|^2}{W_S} x_j(1 - x_j) \right] \end{aligned} \quad (6)$$

20 with normalizing constant

$$\begin{aligned} C(\|\alpha_j\|) &= \int_0^1 x_j^{\theta-1} (1 - x_j)^{\theta-1} \exp \left[-\frac{\|\alpha_j\|^2}{W_S} x_j(1 - x_j) \right] dx_j \\ &= B(\theta, \theta) \cdot {}_1F_1 \left(\theta; \theta + \frac{1}{2}; -\frac{\|\alpha_j\|^2}{4W_S} \right) \end{aligned}$$

21 where ${}_1F_1(a; b; z)$ is Kummer's hypergeometric function. For
22 fixed b and z we have ${}_1F_1(a; b; z) = 1 + \mathcal{O}(a)$ as $a \rightarrow 0$. Hence,
23 as $\theta \rightarrow 0$,

$$p_{\infty}(x_j; \|\alpha_j\|) \propto x_j^{\theta-1} (1 - x_j)^{\theta-1} \exp \left[-\frac{\|\alpha_j\|^2}{W_S} x_j(1 - x_j) \right] \quad (7)$$

24 after suppressing $C(\|\alpha_j\|) \approx B(\theta, \theta)$ which is independent of
25 α_j . A parallel derivation of this stationary law using Wright's
26 formula (Wright 1937; Walsh and Lynch 2018) can be found in
27 Bulmer (1972, eqn. 26).

28 Connecting evolution to mixed-effects models

29 Genomic mixed-effects models assume the conditional distribu-
30 tion of y given G is

$$y = G\beta + \varepsilon \quad (8)$$

31 where y is the focal trait, G is the genotype matrix, and β is the
32 corresponding effect size. Accordingly, the variance decomposition
33 of the trait is

$$\begin{aligned} \text{Var}(y | G) &= \text{Var}(G\beta + \varepsilon | G) \\ &= G\text{Var}(\beta | G)G^T + \text{Var}(\varepsilon | G) \\ &= G\text{Var}(\beta | G)G^T + \text{Var}(\varepsilon) \quad (\because G \perp \varepsilon) \\ &= G\Sigma_{\beta}G^T + \text{Var}(\varepsilon) \end{aligned} \quad (9)$$

34 where $\Sigma_{\beta} = \text{Var}(\beta | G)$.

35 The aforementioned α -model supposes that

$$\Sigma_{\beta} = \text{diag} \left(\frac{h^2}{M} \cdot \frac{1}{[x_j(1 - x_j)]^{\alpha}} \right)_{j=1, \dots, p} \quad (10)$$

36 where M is the number of variants and h^2 is a positive constant
37 often referred to as the single nucleotide polymorphism (SNP)
38 heritability. Here, the effect of G enters the formula through its
39 column-wise frequency x_j . In practice, we only know the sample
40 frequency $\hat{x}_j$ instead of the population frequency x_j , leading to
41 the plug-in estimator

$$\hat{\Sigma}_{\beta} = \text{diag} \left(\frac{h^2}{M} \cdot \frac{1}{[\hat{x}_j(1 - \hat{x}_j)]^{\alpha}} \right)_{j=1, \dots, p} \quad (11)$$

42 replacing Σ_{β} . The $\alpha = 1$ case is the genome-wide complex
43 trait analysis (GCTA) model (Yang *et al.* 2011). Together with G ,
44 $G\hat{\Sigma}_{\beta}G^T$ is a genomic estimate of genetic relatedness, called the
45 genomic genetic relatedness matrix (GRM) (Speed and Balding
46 2014). For a more comprehensive review of different choices of
47 Σ_{β} , see Lehmann *et al.* (2025); Weir and Goudet (2026).

48 Despite the success of equation (11) in data modeling, there
49 are several challenges related to theory and interpretation. First,
50 the form of $\Sigma_{\beta} = \text{Var}(\beta | G)$ in equation (10) is simply assumed
51 *a priori*, so the parameters h^2 and α are merely tuning param-
52 eters without an explicit biological interpretation. α in particular
53 exists in order to heuristically capture the inverse relationship
54 between minor allele frequency and effect size (Zeng *et al.* 2018;
55 Schoech *et al.* 2019). For $\alpha > 0$ there is also the pathology that
56 $[x_j(1 - x_j)]^{-\alpha} \rightarrow \infty$ as $x_j \rightarrow 0$, requiring heuristic GRM parti-
57 tioning over multiple MAF bins to handle rare variants (Yang
58 *et al.* 2015; Wainshtein *et al.* 2022). Finally, substituting $\hat{x}_j$ for x_j
59 can introduce substantial error for rare variants, which tend to
60 exhibit larger sampling variance.

Our framework offers a principled way to overcome these issues. The key is to *deduce* a formula for the SNP-level variance $\Sigma_{\beta,j} = \text{Var}(\beta_j | G)$ from evolutionary principles rather than assume it. Assuming that the mean of β_j is zero, its variance is

$$\begin{aligned} \mathbb{E}[\beta_j^2 | G] &= \mathbb{E}[\mathbb{E}[\beta_j^2 | G, \|\alpha_j\|] | G] \\ &= \mathbb{E}[\mathbb{E}[\beta_j^2 | \|\alpha_j\|] | G] \\ &= \int \mathbb{E}[\beta_j^2 | \|\alpha_j\|] \cdot p(\|\alpha_j\| | G) d\|\alpha_j\| \end{aligned} \quad (12)$$

where α_j is the effect size *vector* of the j -th locus (previously we used the superscript k on α to denote the corresponding k -th trait).

Here, β_j is the effect of locus j on the observed focal trait y appearing in the regression model $y = G\beta + \varepsilon$. By contrast, α_j belongs to the evolutionary model and describes the effect of the same locus on the latent trait z that determines fitness under stabilizing selection in equation (1). These need not be the same quantity because selection may act on an unobserved background trait rather than directly on the focal trait. The two effect sizes are coupled through pleiotropy: a single mutation can perturb both the selected latent trait z and the focal observed trait y .

Fitness effect and its influence on a focal trait

From (12), it remains to determine $\mathbb{E}[\beta_j^2 | \|\alpha_j\|]$, which measures how much the focal trait's effect β_j is determined by the overall fitness effect α_j , and $p(\|\alpha_j\| | G)$, the likely value of overall fitness given the genotype data from a given evolutionary model. We now derive these quantities in the case of stabilizing selection.

For the rest of the paper, we assume

$$\mathbb{E}[\beta_j^2 | \|\alpha_j\|] = \sigma_b^2(1 - \rho_{ab}^2) + \rho_{ab}^2 \cdot \frac{\sigma_b^2}{\sigma_a^2} \cdot \|\alpha_j\|^2 \quad (13)$$

for some constants σ_a^2 , σ_b^2 and ρ_{ab}^2 . σ_a^2 and σ_b^2 control the marginal magnitude of effect sizes α_j and β_j of the latent trait z subject to selection and the observed focal trait y , respectively. The observed trait's effect size exhibits frequency dependence only if β_j is correlated with α_j , and ρ_{ab} precisely controls the strength of this relationship. Equation (13) can be viewed as a generalization of the framework proposed by [Simons et al. \(2018\)](#): with an isotropic FGM,

$$\mathbb{E}[(\alpha_j^1)^2 | \|\alpha_j\|] = \dots = \mathbb{E}[(\alpha_j^d)^2 | \|\alpha_j\|] \quad (14)$$

holds by symmetry (all α_j^k have the same expected magnitude) and

$$\sum_{k=1}^d \mathbb{E}[(\alpha_j^k)^2 | \|\alpha_j\|] = \|\alpha_j\|^2 \quad (15)$$

gives $\mathbb{E}[\beta_j^2 | \|\alpha_j\|] = \|\alpha_j\|^2/d$ where β_j is one of the α_j^k . [Simons et al. \(2018\)](#)'s model is recovered by setting $\rho_{ab} = 1$, $\sigma_b^2/\sigma_a^2 = 1/d$. In this model, β_j is perfectly correlated with $\|\alpha_j\|$ ($\rho_{ab} = 1$).

The original motivation for (13) is a multivariate normal model in which the fitness effect is drawn from

$$\begin{pmatrix} \alpha_j \\ \beta_j \end{pmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{pmatrix} \sigma_a^2 & \rho_{ab}\sigma_a\sigma_b \\ \rho_{ab}\sigma_a\sigma_b & \sigma_b^2 \end{pmatrix} \right) \quad (16)$$

and selection acts on a one-dimensional trait with fitness effect α . Then, $\sigma_a^2 = \mathbb{E}[\|\alpha_j\|^2]$, $\sigma_b^2 = \mathbb{E}[\beta_j^2]$, and $\rho_{ab}^2 = \text{corr}(\|\alpha_j\|^2, \beta_j^2)$ by properties of multivariate normal distributions. An alternative interpretation of equation (13) suggested by a reviewer is that β_j^2 is assigned independently of α_j with probability $1 - \rho_{ab}^2$ and proportionally to $\|\alpha_j\|^2$ with probability ρ_{ab}^2 .

As a reviewer pointed out, writing equation (13) as

$$\mathbb{E}[\beta_j^2 | \|\alpha_j\|] = \sigma_b^2(1 - \rho_{ab}^2) + \rho_{ab}^2 \cdot \frac{\|\alpha_j\|^2}{\sigma_a^2} \cdot \sigma_b^2 \quad (17)$$

reveals that $\|\alpha_j\|^2/\sigma_a^2$ controls the potential direction of β_j because $\mathbb{E}[\|\alpha_j\|^2/\sigma_a^2] = 1$ and ρ_{ab}^2 determines how strongly β_j follows this direction. Finally, σ_b^2 scales this direction to determine the magnitude of β_j .

Approximate genetic architecture

Next we turn to $p(\|\alpha_j\|^2 | G)$. Assuming linkage equilibrium, we have

$$p(\|\alpha_j\|^2 | G) = p(\|\alpha_j\|^2 | G_j) \propto p(G_j | \|\alpha_j\|^2) \cdot p(\|\alpha_j\|^2). \quad (18)$$

The prior $p(\|\alpha_j\|^2)$ is called the distribution of fitness effects (DFE). For the first term on the right, $p(G_j | \|\alpha_j\|^2)$, assume that there are k_j derived alleles observed in G_j . By exchangeability of the $2n$ samples,

$$\begin{aligned} p(G_j | \|\alpha_j\|) &= \int_0^1 p(G_j | x_j) p_\infty(x_j | \|\alpha_j\|) dx_j \\ &= \int_0^1 \binom{2n}{k_j} x_j^{k_j} (1-x_j)^{2n-k_j} p_\infty(x_j | \|\alpha_j\|) dx_j \\ &\propto \int_0^1 \binom{2n}{k_j} x_j^{k_j} (1-x_j)^{2n-k_j} x_j^{\theta-1} (1-x_j)^{\theta-1} \\ &\quad \cdot \exp\left[-\frac{\|\alpha_j\|^2}{W_S} x_j(1-x_j)\right] dx_j \\ &\propto \exp\left[-\frac{\|\alpha_j\|^2}{W_S} \hat{x}_j(1-\hat{x}_j)\right]. \end{aligned} \quad (19)$$

Note that integration over $[0, 1]$ collapses approximately to $x_j \leftarrow \hat{x}_j$ because the beta-binomial distribution degenerates to the sample frequency in large samples $n \rightarrow \infty$ (see Appendix).

For $p(\|\alpha_j\|^2)$, we suppose $\|\alpha_j\|^2$ follows a scaled χ^2 -distribution, with scaling factor σ_a^2 . This is implied if one assumes α_j is normally distributed, as is standard in mixed-effects models. Additionally, the scaled χ^2 -distribution is a particular instance of a Gamma distribution, which is traditionally used to model the DFE ([Eyre-Walker and Keightley 2007](#)). We thus have $\mathbb{E}[\|\alpha_j\|^2] = \sigma_a^2$ and

$$p(\|\alpha_j\|^2 | G_j) \propto \exp\left[-\frac{\|\alpha_j\|^2}{W_S} \hat{x}_j(1-\hat{x}_j)\right] \cdot \frac{1}{\|\alpha_j\|} \exp\left(-\frac{\|\alpha_j\|^2}{2\sigma_a^2}\right),$$

yielding

$$\mathbb{E}[\|\alpha_j\|^2 | G_j] = \frac{\sigma_a^2}{1 + 2\frac{\sigma_a^2}{W_S} \hat{x}_j(1-\hat{x}_j)}. \quad (20)$$

Substituting (13) and (20) into (12), we obtain

$$\mathbb{E}[\beta_j^2 | G_j] = \sigma_b^2 \cdot \left(1 - \rho_{ab}^2 \cdot \frac{2 \frac{\sigma_a^2}{W_S} \hat{x}_j(1 - \hat{x}_j)}{1 + 2 \frac{\sigma_a^2}{W_S} \hat{x}_j(1 - \hat{x}_j)} \right). \quad (21)$$

More generally, formula (21) holds with slightly different constants for any choice of Gamma-distributed prior on the DFE (Appendix).

Framework breakdown

The derivation so far can be broken down into two distinct components that can each be extended on their own. The first component in equation (19) is the forward population genetic model $p(G_j | \|\alpha_j\|)$. This is simply the site frequency spectrum (SFS) conditioned on the latent effect size α_j . Thus, an alternative population-genetic model can be adopted in the framework by replacing this SFS component, for example with a non-equilibrium demography that generalizes the constant-size panmictic assumption.

While the first component reflects the population genetic process conditioned on $\|\alpha_j\|$, equation (20) requires inverting the conditional relationship of $\|\alpha_j\|$ and G_j . According to Bayes' rule, the prior distribution $p(\|\alpha_j\|)$ has to be specified to achieve this goal. We chose the Gamma distribution to obtain a closed-form formula in the main text; but other choices are possible but would require numerically integrating (20).

Simulation and empirical study

SLiM forward simulation

To evaluate the proposed model, we conducted forward simulations using SLiM (Haller *et al.* 2025). Our individual-based simulations assume that stabilizing selection acts according to the fitness function in equation (1). The simulation is based on (16), where a one-dimensional trait is subject to selection while the observed trait is correlated with it through its effect sizes (precise details are given in Appendix). Notably, this simulation setup differs from most other LMM benchmarking studies, which typically condition on fixed genotypes and only vary the random effects and the residual noise of the regression model (Speed *et al.* 2017; Gazal *et al.* 2019; Privé *et al.* 2020). The genic variance attributed to variant j was computed as $\mathbb{E}[\beta_j^2 | \hat{x}_j] \cdot 2\hat{x}_j(1 - \hat{x}_j)$. The total genic variance was then obtained by summing over all j . Note that this is the numerator of the usual SNP-heritability h^2 . The minor allele frequency (MAF) bins for the MAF-stratified α -model were [0,0.001], (0.001,0.01], (0.01, 0.1], (0.1,0.2], (0.2,0.3], (0.3,0.4], (0.4,0.5] (Lee *et al.* 2013).

Variance component estimation

We evaluated equation (21) by simulating traits with small non-genetic noise σ_e^2 . For each replicate, we fit the evolutionary mixed model to the full genotype matrix after removing monomorphic sites and computing the sample allele frequencies $\hat{x}_j$ from the remaining dosage columns. These frequencies were inserted into the variance formula in equation (21), and the phenotype was analyzed after adding Gaussian noise of fixed variance $\sigma_e^2 = 10$. With ρ_{ab} fixed to its simulation value, we then maximized the marginal likelihood over the two log-transformed parameters $\log(\sigma_a^2/W_S)$ and the overall prior scale parameter $\log(\sigma_b^2)$ using an L-BFGS-B optimizer implemented in *scipy* (Virtanen *et al.* 2020).

We first examined whether REML recovers the variance components implied by equation (21). Figure 1 shows that the fitted values preserve the ordering of the true parameter combinations across the simulation grid, but they lie systematically below the naive reference σ_a^2/W_S . The discrepancy is most visible when selection-induced frequency dependence is strongest (larger σ_a^2/W_S). At the same time, the estimates are substantially closer to the Bulmer-corrected reference lines than to the uncorrected target, indicating that the main departure from the single-locus theory is consistent with attenuation from linkage disequilibrium rather than a failure of the functional form itself. See Appendix for additional details about the correction. The $\rho_{ab}^2 = 0$ panels are uninformative, because selection on the latent trait does not induce frequency dependence in the focal trait when the two traits are uncorrelated. Formally, one can see in equation (21) that the frequency-dependent term entirely vanishes when $\rho_{ab}^2 = 0$.

The overall scale parameter σ_b^2 is considerably easier to recover. Figure 2 shows that REML estimates of σ_b^2 remain centered near the true value across the full grid of W_S and ρ_{ab}^2 , with only modest replicate-to-replicate variability. This further supports the robustness of the functional form of equation (21). Taken together, these results suggest that the focal-trait variance is well identified, whereas the selection-sensitive component is recovered chiefly through the combined quantity σ_a^2/W_S and is therefore more sensitive to departures from the single-locus approximation.

Genetic prediction

For the prediction experiments, we first added Gaussian noise of variance $\sigma_e^2 = 100$ to each replicate phenotype, then split individuals into equally sized training and validation sets using the replicate identifier as the random seed. All fits were carried out on the training set after centering by the training mean. Under the evolutionary model, we jointly estimated σ_a^2/W_S , the overall variance scale σ_b^2 , and σ_e^2 by REML. For the α -model baselines with $\alpha \in \{0, 0.5, 1\}$, we instead fit the variance scale and σ_e^2 by REML using per-variant prior variances proportional to $[\hat{x}_j(1 - \hat{x}_j)]^{-\alpha}$. The fitted prior variances were then used to form posterior mean BLUP effect estimates on the training set, which were evaluated on the held-out individuals by R^2 .

We next compared held-out BLUP accuracy under the evolutionary model and three α -model baselines. Figure 3 reports the validation R^2 across the same grid of simulation parameters. The strongest pattern is the poor performance of the $\alpha = 1$ model, which underperforms substantially in every regime. The $\alpha = 0.5$ model is closer, but still remains below the evolutionary fit throughout most of the grid. By contrast, the evolutionary model and the $\alpha = 0$ model are often quite similar, especially when $\rho_{ab}^2 = 0$, where equation (21) reduces to an approximately frequency-independent architecture. When the coupling between the focal and selected traits is stronger, the evolutionary model typically retains a small but consistent advantage.

The distinction is sharper when one examines the variance components obtained from the prediction fits. In Figure 4, the evolutionary model remains close to the true σ_b^2 throughout the simulation grid. The $\alpha = 0$ model is approximately unbiased only when $\rho_{ab}^2 = 0$ and becomes downward biased once the focal trait inherits frequency dependence from the selected trait. The degradation is more pronounced in smaller values of W_S corresponding to stronger selection. The $\alpha = 0.5$ and $\alpha = 1$ fits underestimate σ_b^2 much more severely in all scenarios. Hence,

the similarity in predictive accuracy between the evolutionary model and $\alpha = 0$ in some panels should not be interpreted as recovery of the same architecture: the heuristic model can predict reasonably well while still mischaracterizing the underlying variance components. Overall, this shows that the α -model fails to capture the frequency-dependent genetic architecture properly.

Figures A1 and A2 repeat the same training/validation pipeline but fix the evolutionary fit at $\rho_{ab}^2 = 1$. Held-out prediction is nearly unchanged relative to Figure 3, and estimation of σ_b^2 remains close to the full model, although with somewhat greater dispersion in some settings. This supports the reduced parameterization as a useful approximation when separate identification of ρ_{ab}^2 and σ_a^2/W_S is weak.

Frequency stratification in α -model

The α -model baselines considered so far perform substantially worse than the evolutionary model, which seems at odds with the empirical success of the α -model in the literature. Here we show that much of this success can be explained by the minor allele frequency (MAF) stratification used in many modern implementations (Lee et al. 2013; Yang et al. 2015; Gazal et al. 2017; Speed et al. 2017). The MAF-stratified, or binned, α -model fits

$$\mathbb{E}[\beta^2 | x] = \frac{h_{i,\text{bin}}^2}{M} \cdot \frac{1}{[x(1-x)]^\alpha} \quad \text{for } x \in [x_{i-1}, x_i] \quad (22)$$

with MAF bins $0 = x_1 < \dots < x_b = 1$, bin-specific $h_{i,\text{bin}}^2$, and a shared α across bins. Because each curve (22) only needs to describe a narrow MAF interval, the stratified model is much more flexible than the vanilla α -model in capturing the relationship between $\mathbb{E}[\beta^2 | x]$ and allele frequency. Indeed, with sufficiently fine bins, this piecewise construction can approximate an arbitrary frequency-dependent curve, since over a short enough segment any reasonably smooth curve becomes nearly indistinguishable.

Figure 5 illustrates this point under stabilizing selection: the stratified α -model tracks the evolutionary curve for $\mathbb{E}[\beta^2 | x]$ far better than the vanilla α -model. The agreement is especially strong in the higher-MAF bins, which account for most of the genic variance of the trait (Figure 6). This closer match translates into substantially better BLUP performance than the vanilla α -model (Figure 7). In other words, the empirical success of the α -model is largely driven by MAF binning, which allows it to mimic the true evolutionary frequency dependence. Even if the α -model is a poor fit over the MAF spectrum globally, confining it to short segments largely salvages the model.

Even with that added flexibility, however, the main advantages of the framework developed here remain twofold. First, the evolutionary model is statistically more parsimonious: the stratified α -model requires one additional $h_{i,\text{bin}}^2$ parameter for every extra MAF bin. Second, the evolutionary model is more interpretable, because it estimates the raw mutational scale σ_b^2 directly. By contrast, the stratified model's $h_{i,\text{bin}}^2$ parameters are best viewed as tuning knobs chosen to match the binned curve to the underlying evolutionary relationship (Figures 8, A7, and A8).

Discussion

We presented a general procedure for deriving evolution-aware mixed-effects models and applied it to stabilizing selection with pleiotropy. This yields a frequency-dependent effect size formula

expressed directly in terms of evolutionary parameters, while still allowing those parameters to be estimated by restricted maximum likelihood (REML). In that sense, the framework links an interpretable fitness landscape model to standard statistical genetic practice: the same model can be used both to learn about the evolutionary architecture of a trait and to perform downstream tasks such as genetic prediction by best linear unbiased prediction (BLUP). In forward simulations, this evolutionary model outperformed the conventional α -model across the values of α we considered, which supports the practical value of deriving the frequency dependence from an explicit evolutionary model rather than imposing it heuristically.

A central ingredient in this construction is the model of pleiotropy. Empirical studies suggest that pleiotropy is pervasive in human complex traits, and Simons et al. (2018) captured this idea within Fisher's geometric model (FGM) by assuming isotropic mutational effects. Under that assumption, each axis of the underlying fitness landscape contributes equally, so the effect of a mutation on the focal trait is spread symmetrically across directions in trait space. Equation (13) generalizes this relationship by allowing the focal trait to be coupled to overall fitness in a potentially non-isotropic way. One concrete example is a jointly Gaussian model for mutational effects, which has been studied in earlier theoretical work and which we also used in our simulations (Kimura and Crow 1964; Bürger 1986, 2000; Walsh and Lynch 2018). More broadly, this suggests that the present framework should not be tied to a single pleiotropy mechanism: an important next step is to investigate what other biological models lead to the same or similar relationship in equation (13).

The simulations also clarify what aspects of the model are practically identifiable. In particular, jointly estimating ρ_{ab} and σ_a^2/W_S is difficult because $\hat{x}_j(1 - \hat{x}_j)$ is typically both small and weakly variable under the U-shaped allele-frequency spectrum. As a result, the denominator in equation (21) is usually close to one, so the data mainly identify the product $\rho_{ab}^2 \sigma_a^2/W_S$ rather than its two components separately. This is why fixing $\rho_{ab} = 1$ and interpreting the fitted quantity as $\rho_{ab}^2 \sigma_a^2/W_S$ works reasonably well in practice. As we show in Figures A1 and A2, this reduced parameterization lowers precision somewhat but has little effect on BLUP accuracy, and estimation of σ_b^2 remains stable. That asymmetry is itself informative: σ_b^2 controls the overall scale of the covariance structure, whereas σ_a^2/W_S enters only through a weaker modulation by allele frequency. Even when the evolutionary decomposition is only partially identified, the model can therefore still recover the aggregate variance carried by the focal trait.

This perspective also helps explain the behavior of the α -model baselines. When $\rho_{ab}^2 = 0$, the focal trait is effectively decoupled from the selected background trait, so stabilizing selection induces essentially no frequency dependence and the model collapses to an approximately frequency-independent prior. That regime is therefore a useful sanity check: the corresponding panels are uninformative for σ_a^2/W_S , and the evolutionary fit and the $\alpha = 0$ baseline give nearly identical predictions. More generally, because most variants are rare, one often has $\mathbb{E}[\beta_j^2 | \hat{x}_j] \approx \mathbb{E}[\beta_j^2]$, so the genetic relatedness matrices produced by the evolutionary model and the $\alpha = 0$ model differ mainly by an overall scale factor to which BLUP is largely insensitive. Nevertheless, that similarity in prediction should not be mistaken for recovery of the same architecture. Once the focal trait is coupled to the selected background, the $\alpha = 0$ model

absorbs the missing frequency dependence by pushing its estimate of σ_b^2 downward, whereas the nonzero α -models degrade more substantially in both prediction and variance-component estimation. Furthermore, the α -model fails to correctly recover the genic variance across MAF bins, substantially exaggerating the contribution of rare variants (Figures A9, A10, and A11). The comparison therefore shows that predictive performance alone can obscure architectural misspecification, and that an evolutionarily derived frequency-dependence model is preferable to purely statistical heuristics.

The frequency-stratified α -model helps reconcile this conclusion with the empirical success of α -based methods. A single global α -model curve is too rigid to match the stabilizing selection relationship over the full MAF range, but once the spectrum is partitioned into narrow bins, each local fit only has to approximate the target over a short interval where the mismatch is much less pronounced. This repair is especially effective because most of the genic variance is carried by the higher-frequency bins, where the binned approximation is accurate, so BLUP accuracy can improve substantially even when the global functional form is still misspecified. The evolutionary model nevertheless retains an important advantage because it captures the same frequency dependence with one mechanistic formula rather than a collection of bin-specific tuning parameters. In that sense, the binned α -model succeeds by flexible local approximation, whereas the evolutionary model remains both more parsimonious and more directly interpretable.

We did not carry out a real-data analysis in this study, and closing this gap will require an efficient implementation of the proposed variance-component model for modern genomic datasets. In the meantime, we can draw some real-world implications from the new model. First, the evolutionary model can isolate the mutational variance σ_b^2 , which measures the magnitude of new mutations, from existing population data rather than costly lab experiments (Lynch and Walsh 1998; Lee *et al.* 2026). For genic variance, the evolutionary model is unlikely to produce estimates that differ substantially from the α -model, because existing methods implement the MAF-stratified α -model, which approximates the evolutionary model well as shown in Figures A5, A6, and 6. Nevertheless, the evolutionary model offers a more parsimonious parameterization for mixed-effects models that is likely more resource-efficient and numerically stable.

Finally, these results should be interpreted in light of the simplifying assumptions used to obtain a tractable formula. Our derivation relies on genotype-likelihood factorization in the spirit of composite likelihood, stationarity of the population, a low mutation rate, and weak selection relative to recombination. In the full multilocus setting, the dynamics of a focal locus depend on the genetic background through linkage, even in the strictly additive setting considered here, and ignoring that dependence lets us reduce the analysis to a single locus. We suspect that this omission is responsible for the systematic underestimation of σ_a^2/W_S seen in the simulations. In particular, linkage disequilibrium induces the Bulmer effect, which creates background forces opposing selection and thereby weakens the effective selective pressure acting on a locus. Negm and Veller (2026) formalized this reduction as a multiplicative factor, which algebraically has the same effect as increasing the selection width W_S and hence decreasing σ_a^2/W_S . By expanding their formula with further approximations, we found that the Bulmer effect quantitatively predicts the observed underestimation;

Appendix gives the details. Extending the framework to treat linkage disequilibrium and richer models of pleiotropy more explicitly is therefore the most natural direction for future work.

Data availability

The codes and scripts used in this paper to conduct simulations and statistical analysis have been deposited in https://github.com/hanbin973/param_arch_stab_paper. Real data is not applicable.

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Conflicts of interest

Not applicable.

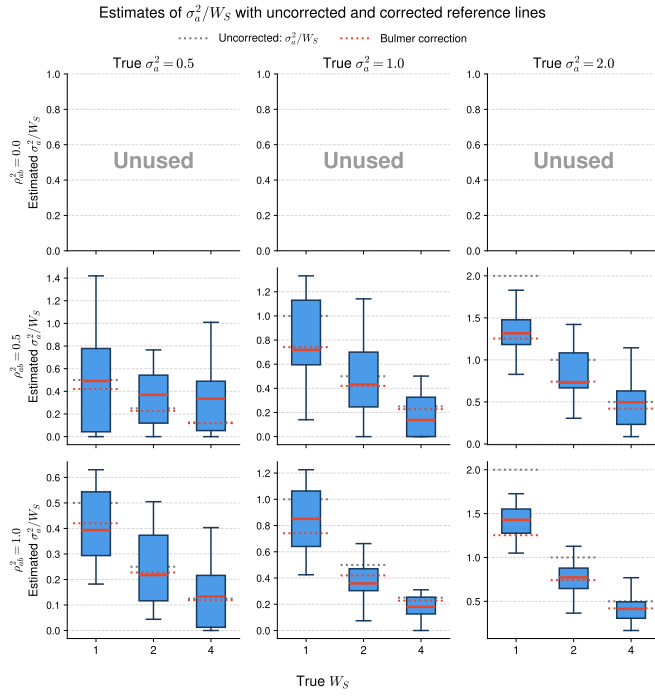


Figure 1 REML estimates of σ_a^2 / W_S across forward-simulation replicates. Columns correspond to the true σ_a^2 , rows correspond to the true ρ_{ab}^2 , and the x-axis gives the true W_S . Gray dotted lines show the uncorrected target σ_a^2 / W_S , while red dotted lines show the Bulmer-corrected reference derived in the appendix. The $\rho_{ab}^2 = 0$ panels are unused because the focal trait then carries no information about selection on the latent trait.

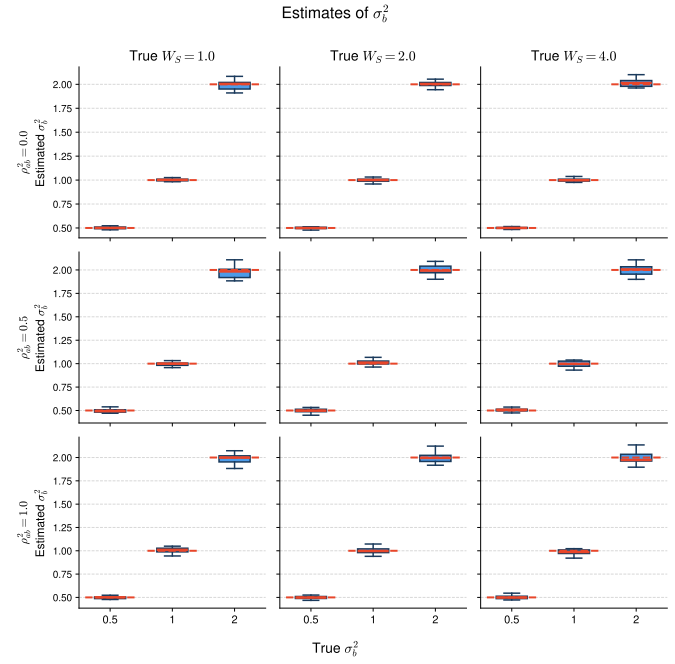


Figure 2 REML estimates of σ_b^2 across forward-simulation replicates. Columns correspond to the true W_S , rows correspond to the true ρ_{ab}^2 , and the x-axis gives the true σ_b^2 . Horizontal reference lines indicate the true values.

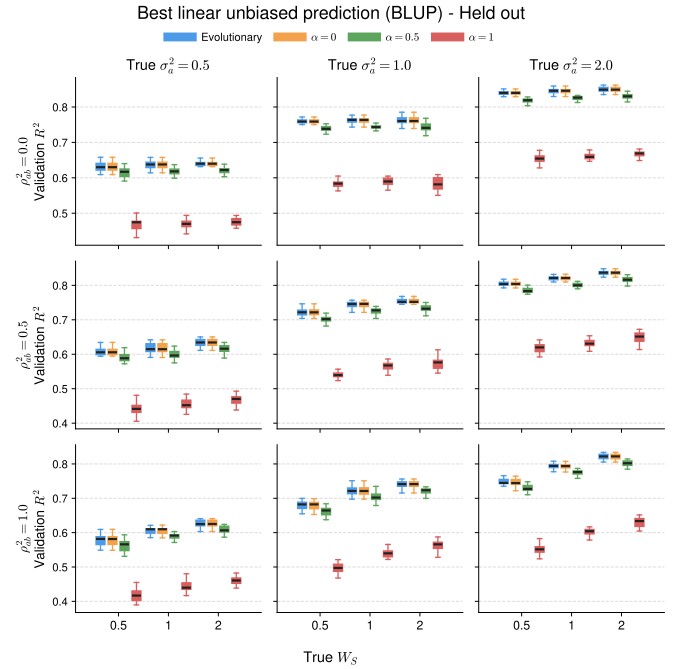


Figure 3 Held-out BLUP validation R^2 for the evolutionary model and the α -model baselines. Columns correspond to the true σ_a^2 , rows correspond to the true ρ_{ab}^2 , and the x-axis gives the true W_S . Error bars summarize variation across replicate populations.

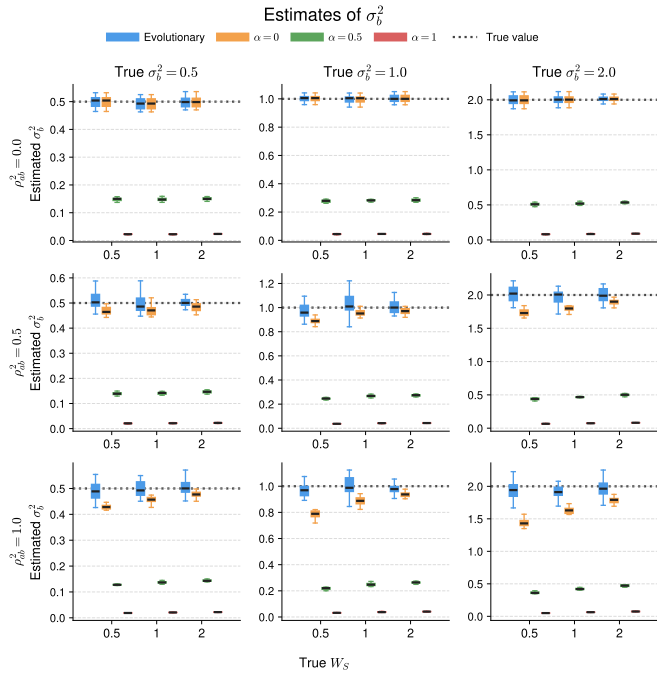


Figure 4 Estimates of σ_b^2 obtained from the BLUP fits. Columns correspond to the true σ_b^2 , rows correspond to the true ρ_{ab}^2 , and the x-axis gives the true W_S . The dotted horizontal line marks the true value.

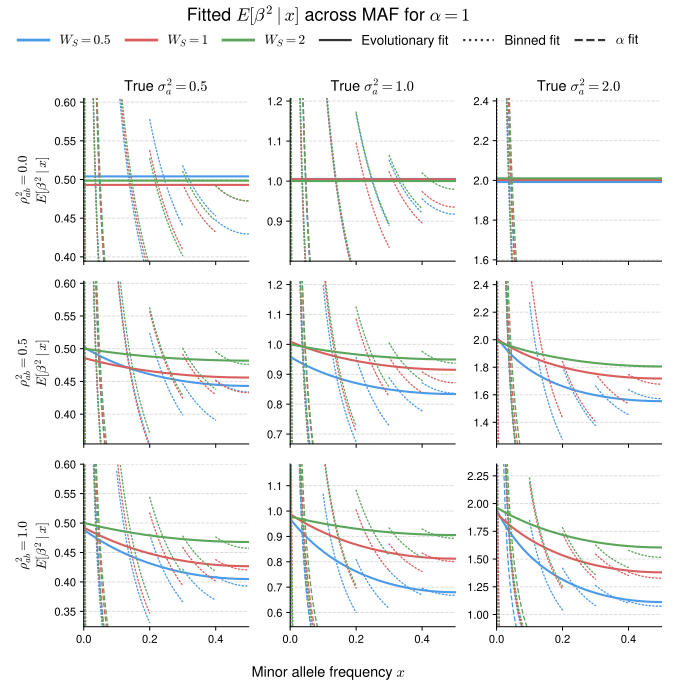


Figure 5 Relationship between allele frequency x and $E[\beta^2 | x]$ with $\alpha = 1$. The binned model closely tracks the evolutionary curve over the MAF range shown. Corresponding $\alpha = 0$ and $\alpha = 0.5$ versions are given in Figures A3 and A4. The solid, dotted, and dashed lines correspond to the evolutionary, binned α , and vanilla α models, respectively. The curves were drawn from the median parameter estimates across 20 replicates.

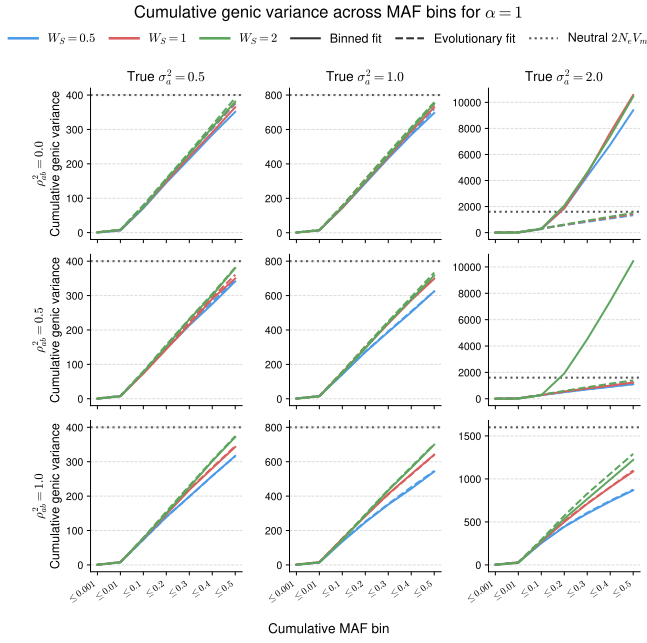


Figure 6 Cumulative genetic variance across MAF bins computed from $\alpha = 1$ -model with MAF stratification. Most of the variance is concentrated in the higher-frequency bins where the binned approximation is strongest. Corresponding $\alpha = 0$ and $\alpha = 0.5$ versions are given in Figures A5 and A6. The solid and dashed lines correspond to the binned α and evolutionary models, respectively. The dotted line is the expected genetic variance under neutrality, serving as the upper bound. The curves were drawn from the median parameters estimates across 20 replicates.

Best linear unbiased prediction (BLUP) - Held out, MAF-binned fits

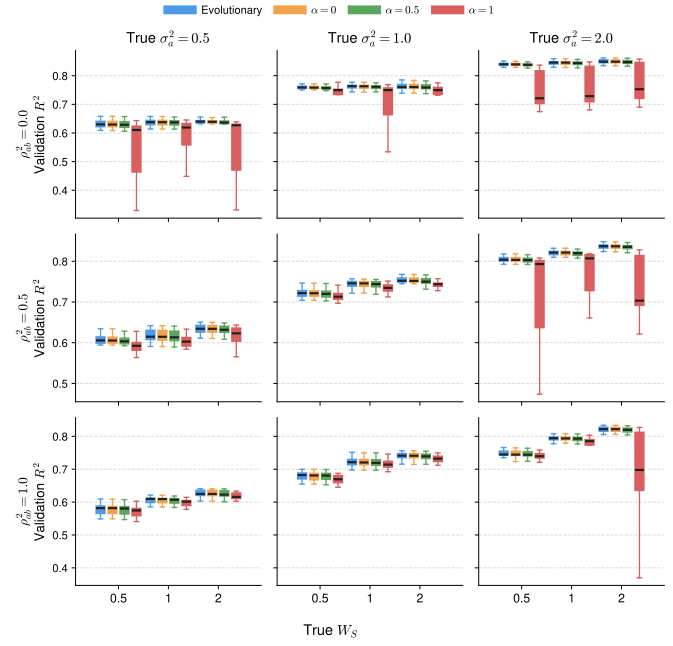


Figure 7 Held-out BLUP validation R^2 under the frequency-stratified α -model. MAF binning substantially improves prediction relative to the vanilla α -model by allowing a closer approximation to the evolutionary frequency dependence.

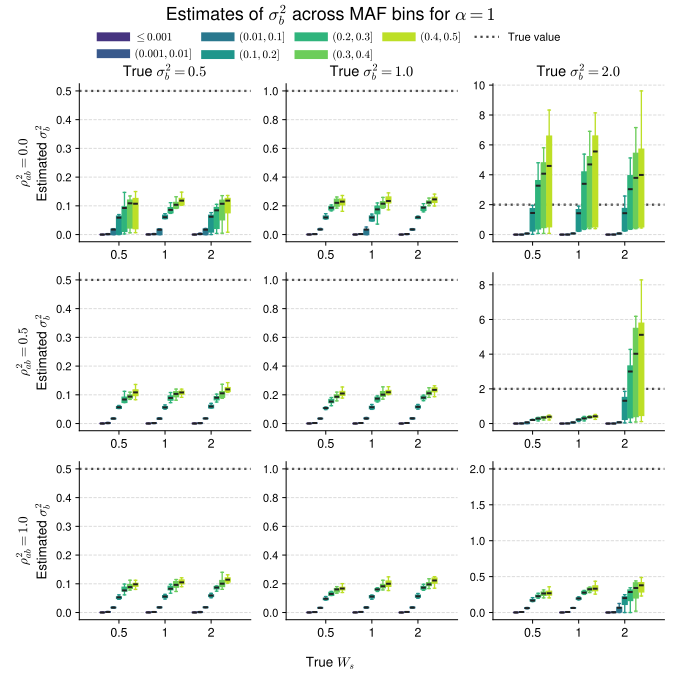


Figure 8 Bin-specific variance parameters with $\alpha = 1$. These parameters act as tuning parameters that adapt the stratified curve to the underlying evolutionary relationship. Corresponding $\alpha = 0$ and $\alpha = 0.5$ versions are given in Figures A7 and A8.

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Appendix

Mean-field approach to linkage disequilibrium

So far, we have completely ignored linkage, which creates theoretical challenges. To analyze the dynamics of an individual locus in general, we must account for its many random interactions with other loci. To simplify the problem, we adopt the *mean-field* approximation, in which the locus is assumed to interact with the rest of the system in a simplified fashion. Here, the field is the rest of the genome, or the genetic background, with which the focal locus interacts. Because this background consists of a huge number of loci and is stochastic in principle, we instead treat it as a deterministic *average*, which gives rise to the term mean-field.

We work with each locus while leaving all else as a deterministic aggregate. Under this framework, one can further refine equation (2) as (after scaling by $2N$)

$$s_{\text{eff}}(x_j) = \left(x_j - \frac{1}{2}\right) \cdot \left(1 - \frac{1}{2\bar{c}_j} \frac{V_g}{V_S}\right)^2 \cdot \frac{\alpha_j^2}{W_S} \quad (\text{A1})$$

where $\bar{c}_j$ and $\bar{c}$ are local and global harmonic means of the recombination fractions, respectively, given by

$$\bar{c}_j = (L-1) \left/ \sum_{k:k \neq j} \frac{1}{c_{jk}} \right. \quad \text{and} \quad \bar{c} = L(L-1) \left/ \sum_{j,k:j \neq k} \frac{1}{c_{jk}} \right. \quad (\text{A2})$$

and $V_g = 2 \sum_j x_j(1-x_j)\alpha_j^2$ is the genic variance of z as derived in [Negm and Veller \(2026\)](#). For simplicity, let u_j be the normalized position of locus j so that Lu_j is the position in base pairs, and write $h(u_j) = 1/2\bar{c}_j$.

The difficulty arising in equation (A1) is that V_g is also a random variable because it is a function of the random variables x_j . We take another round of mean-field approximation so that $V_g = 2L \cdot \mathbb{E}[x(1-x)\alpha^2]$ is a constant. By further assuming a uniform mutation rate, or equivalently rescaling the genome, we arrive at

$$\begin{aligned} \alpha &\sim N(0, \sigma_a^2) \\ u &\sim \text{Uniform}[1/L, (L-1)/L] \\ x | \alpha, u &\sim x^{\theta-1}(1-x)^{\theta-1} \exp \left[- \left(1 - h(u) \cdot \frac{V_g}{V_S}\right)^2 \cdot \frac{\alpha^2}{W_S} x(1-x) \right] \\ V_g &= 2L \cdot \mathbb{E}[x(1-x)\alpha^2]. \end{aligned} \quad (\text{A3})$$

under stationarity as before. Solving for V_g with $h(u)V_g \ll V_S$ gives

$$V_g = \frac{2L\theta\sigma_a^2}{2 + \frac{\sigma_a^2}{W_S}}. \quad (\text{A4})$$

which shows that the shape of $h(u)$ is irrelevant under the $V_g \ll V_S$ approximation. A locus-level equivalent is derived in [Keightley and Hill \(1988\)](#). In the limit of diminishing selection, with $W_S \rightarrow \infty$, the formula reduces to the classic neutral result $V_G = L\theta\sigma_a^2 = 4NL\mu\sigma_a^2$ ([Lynch and Hill 1986](#)).

We now follow the same steps as before to deduce

$$\begin{aligned} \mathbb{E}[\beta_j^2 | \hat{x}_j, u_j] &= \sigma_{b,j}^2 \cdot \left[1 - \rho_{ab,j}^2 \cdot \frac{2 \left(1 - h(u_j) \frac{V_g}{V_S}\right)^2 \frac{\sigma_{a,j}^2}{W_S} \hat{x}_j(1-\hat{x}_j)}{1 + 2 \left(1 - h(u_j) \frac{V_g}{V_S}\right)^2 \frac{\sigma_{a,j}^2}{W_S} \hat{x}_j(1-\hat{x}_j)} \right] \end{aligned} \quad (\text{A5})$$

which reduces to

$$\mathbb{E}[\beta_j^2 | \hat{x}_j, u_j] = \frac{\sigma_{b,j}^2}{1 + 2 \left(1 - h(u_j) \frac{V_g}{V_S}\right)^2 \frac{\sigma_{a,j}^2}{W_S} \hat{x}_j(1-\hat{x}_j)} \quad (\text{A6})$$

if $\rho_{ab} = 1$, a perfect correlation between α and β . As before, if $\rho_{ab} = 0$, it simply becomes $\sigma_{b,j}^2$ because selection on z has no influence on y at all. For the correction factors in Figure 1, we used the global harmonic mean of the recombination fraction.

Simulation details

The population followed a panmictic demographic history with a constant diploid population size of $N = 10,000$ individuals. Each genome was of length $L = 2 \times 10^6$ with mutation rate of $\mu = 10^{-8}$ and recombination rate of $r = 10^{-6}$. The configuration was borrowed from [Hayward and Sella \(2022\)](#) to set $L\mu = 0.02$ mutations per genome per generation, $2NL\mu = 400 \gg 1$ to ensure polygenicity, and $r/\mu = 10^2$ for low linkage disequilibrium (see Appendix). The mutational effect sizes α were simulated with $\sigma_a^2 \in \{0.5, 1.0, 2.0\}$ and selection width $W_S = V_S/2N = \{1.0, 2.0, 4.0\}$. As V_S is of scale $\mathcal{O}(N)$, the environmental variance V_E and the genetic variance V_G were negligible. Hence, the fitness function (1) was used in the simulations. For each pair of σ_a^2 and W_S , a single burn-in of $10N$ generations was followed by 20 separate replicates of $5N$ generations.

To simulate pleiotropic selection, after a total of $15N$ generations we drew the observed effect sizes β conditional on α according to the joint distribution in equation (16). This is equivalent to sampling β after the SLiM simulation because β has no effect on the simulated dynamics; only α determines selection. For σ_b^2 and ρ_{ab}^2 , we chose them to be the same as σ_a^2 and to take values in the grid $\{0.00, 0.50, 1.00\}$, respectively.

Large sample asymptotics for Beta-Binomial distribution

We elaborate on the large n asymptotics used in equation (19). After absorbing all terms that do not depend on $\|\alpha_j\|$ into the proportionality constant, the integral in (19) becomes

$$\int_0^1 x_j^{k_j+\theta-1} (1-x_j)^{2n-k_j+\theta-1} \exp \left[- \frac{\|\alpha_j\|^2}{W_S} x_j(1-x_j) \right] dx_j. \quad (\text{A7})$$

The polynomial factor can be normalized into a Beta density, yielding

$$\begin{aligned} &\int_0^1 x_j^{k_j+\theta-1} (1-x_j)^{2n-k_j+\theta-1} \exp \left[- \frac{\|\alpha_j\|^2}{W_S} x_j(1-x_j) \right] dx_j \\ &= \text{B}(k_j + \theta, 2n - k_j + \theta) \cdot \mathbb{E} \left[\exp \left(- \frac{\|\alpha_j\|^2}{W_S} X_n(1-X_n) \right) \right] \end{aligned} \quad (\text{A8})$$

where

$$X_n \sim \text{Beta}(k_j + \theta, 2n - k_j + \theta). \quad (\text{A9})$$

The mean and variance of X_n are

$$\mathbb{E}[X_n] = \frac{k_j + \theta}{2n + 2\theta}, \quad \text{Var}(X_n) = \frac{(k_j + \theta)(2n - k_j + \theta)}{(2n + 2\theta)^2(2n + 2\theta + 1)}. \quad (\text{A10})$$

1 Writing $\hat{x}_j = k_j / (2n)$, we obtain

$$\mathbb{E}[X_n] = \hat{x}_j + \mathcal{O}(n^{-1}), \quad \text{Var}(X_n) = \mathcal{O}(n^{-1}). \quad (\text{A11})$$

2 Hence, if $k_j / (2n) \rightarrow x^*$, then Chebyshev's inequality implies
 3 that $X_n \xrightarrow{P} x^*$. In the notation of the main text, this means that
 4 the Beta distribution concentrates around the observed sample
 5 frequency $\hat{x}_j$ as $n \rightarrow \infty$.

6 Now define

$$g(x) = \exp \left[-\frac{\|\alpha_j\|^2}{W_S} x(1-x) \right]. \quad (\text{A12})$$

7 Since g is continuous on $[0, 1]$, the continuous mapping theorem
 8 gives

$$g(X_n) \xrightarrow{P} \exp \left[-\frac{\|\alpha_j\|^2}{W_S} \hat{x}_j(1 - \hat{x}_j) \right]. \quad (\text{A13})$$

9 Moreover, $0 \leq g(X_n) \leq 1$, so bounded convergence implies

$$\mathbb{E}[g(X_n)] \rightarrow \exp \left[-\frac{\|\alpha_j\|^2}{W_S} \hat{x}_j(1 - \hat{x}_j) \right]. \quad (\text{A14})$$

10 Therefore,

$$p(G_j \mid \|\alpha_j\|) \propto \exp \left[-\frac{\|\alpha_j\|^2}{W_S} \hat{x}_j(1 - \hat{x}_j) \right] \quad (\text{A15})$$

11 for large n , where the omitted factor $\binom{2n}{k_j} B(k_j + \theta, 2n - k_j + \theta)$
 12 does not depend on $\|\alpha_j\|$ and is absorbed into the proportionality
 13 constant.

14 Genetic architecture under Gamma DFE

15 Suppose that $\|\alpha_j\|^2$ follows a Gamma distribution

$$p(\|\alpha_j\|^2) \propto \|\alpha_j\|^{2(k-1)} \exp \left(-\frac{k\|\alpha_j\|^2}{\sigma_a^2} \right)$$

16 which reduces to the σ_a^2 -scaled χ^2 -distribution of the main text
 17 when $k = 1/2$. Then, the posterior distribution

$$\begin{aligned} p(\|\alpha_j\|^2 \mid G_j) \\ \propto \exp \left[-\frac{\|\alpha_j\|^2}{W_S} \hat{x}_j(1 - \hat{x}_j) \right] \cdot \|\alpha_j\|^{2(k-1)} \exp \left(-\frac{k\|\alpha_j\|^2}{\sigma_a^2} \right) \end{aligned}$$

18 is also a Gamma distribution with mean

$$\mathbb{E}[\|\alpha_j\|^2 \mid G_j] = \frac{\sigma_a^2}{1 + \frac{\sigma_a^2}{kW_S} \hat{x}_j(1 - \hat{x}_j)}. \quad (\text{A16})$$

19 The conditional mean of β_j^2 is therefore

$$\begin{aligned} \mathbb{E}[\beta_j^2 \mid G_j] &= \sigma_b^2(1 - \rho_{ab}^2) + \rho_{ab}^2 \cdot \frac{\sigma_b^2}{\sigma_a^2} \cdot \mathbb{E}[\|\alpha_j\|^2 \mid G_j] \\ &= \sigma_b^2(1 - \rho_{ab}^2) + \rho_{ab}^2 \cdot \frac{\sigma_b^2}{\sigma_a^2} \cdot \frac{\sigma_a^2}{1 + \frac{\sigma_a^2}{kW_S} \hat{x}_j(1 - \hat{x}_j)} \\ &= \sigma_b^2 \cdot \left(1 - \rho_{ab}^2 \cdot \frac{\frac{\sigma_a^2}{kW_S} \hat{x}_j(1 - \hat{x}_j)}{1 + \frac{\sigma_a^2}{kW_S} \hat{x}_j(1 - \hat{x}_j)} \right) \end{aligned} \quad (\text{A17})$$

20 which reduces to equation (21) with $k = 1/2$. Hence, the func-
 21 tional form for $\mathbb{E}[\beta_j^2 \mid G_j]$ stays the same under Gamma DFE.

22 Note that k controls how clustered the distribution is around
 23 0. For small k , the mass of $p(\|\alpha_j\|^2)$ is concentrated near 0;
 24 whereas for large k , the probability mass shifts away from zero.
 25 For instance, if $k \rightarrow \infty$, meaning that large effect variants are
 26 common, equation (A17) reduces to

$$\mathbb{E}[\beta_j^2 \mid G_j] = \sigma_b^2. \quad (\text{A18})$$

27 Therefore, the frequency dependence is stronger when large
 28 effect variants are rarer and vice versa.

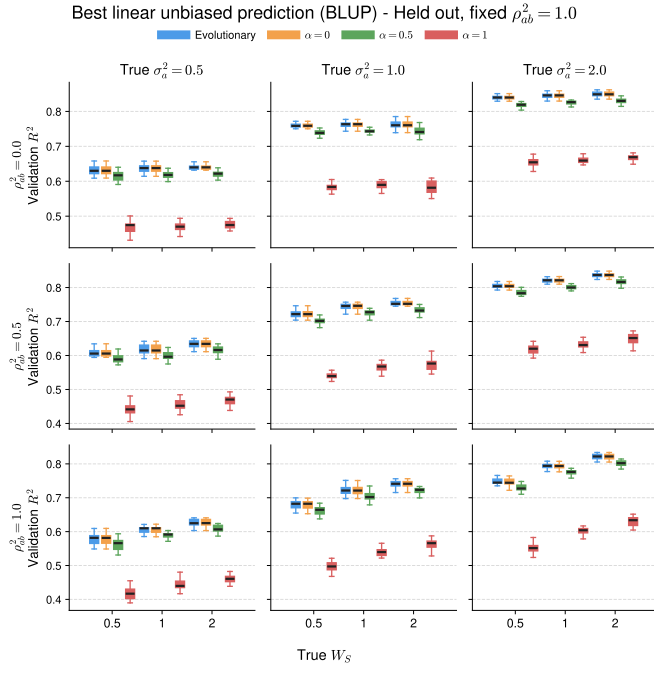


Figure A1 Held-out BLUP validation R^2 when the evolutionary model is fit with ρ_{ab}^2 fixed to 1. Columns correspond to the true σ_a^2 , rows correspond to the true ρ_{ab}^2 , and the x-axis gives the true W_S . Error bars summarize variation across replicate populations.

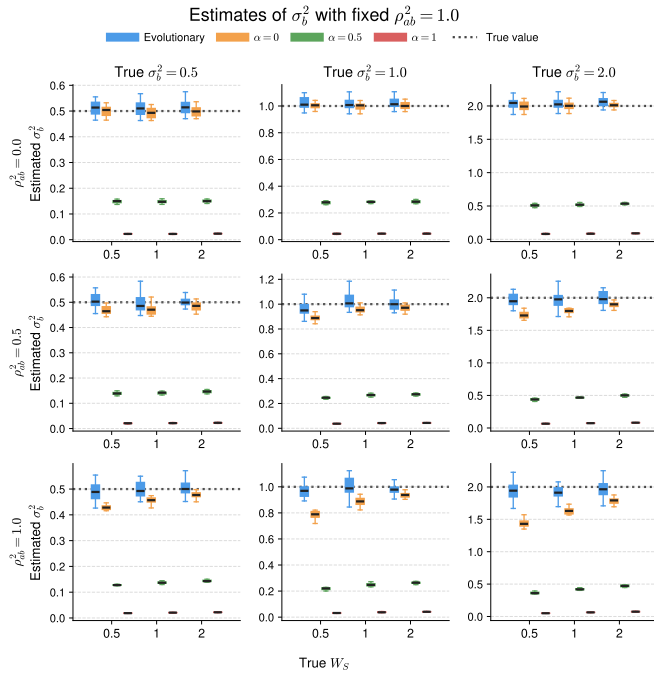


Figure A2 Estimates of σ_b^2 when the evolutionary model is fit with ρ_{ab}^2 fixed to 1. Columns correspond to the true σ_b^2 , rows correspond to the true ρ_{ab}^2 , and the x-axis gives the true W_S . The dotted horizontal line marks the true value.

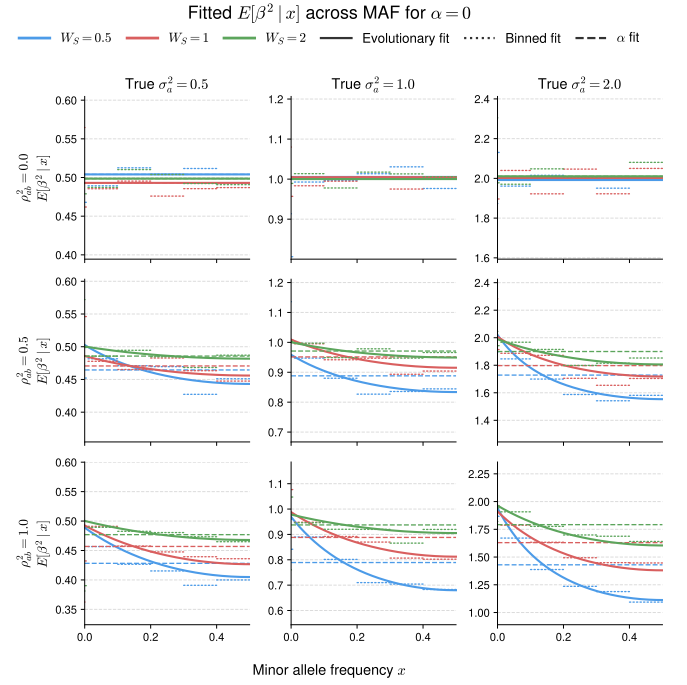


Figure A3 Relationship between allele frequency and $E[\beta^2 | x]$ for $\alpha = 0$.

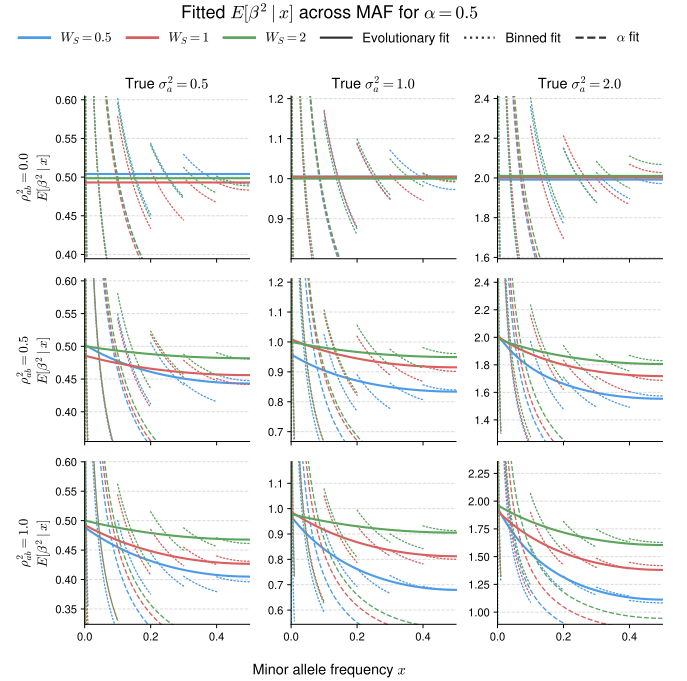


Figure A4 Relationship between allele frequency and $E[\beta^2 | x]$ for $\alpha = 0.5$.

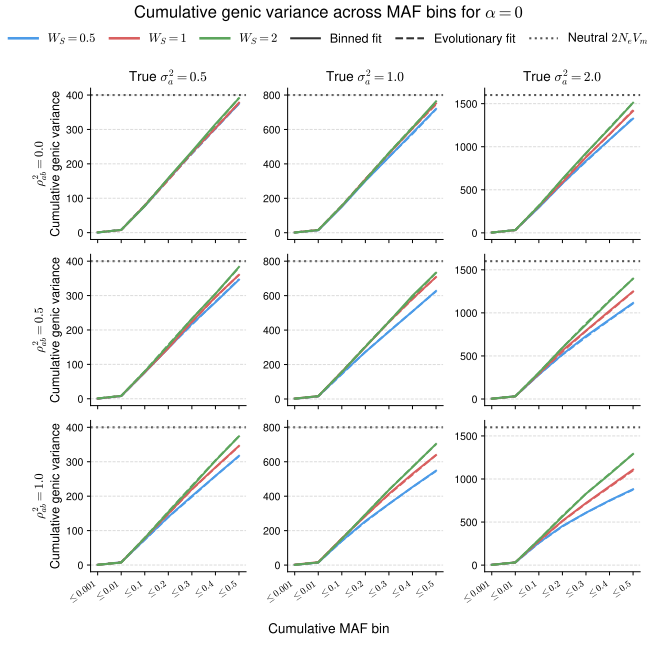


Figure A5 Cumulative genetic variance across MAF bins computed from $\alpha = 0$ -model with MAF stratification.

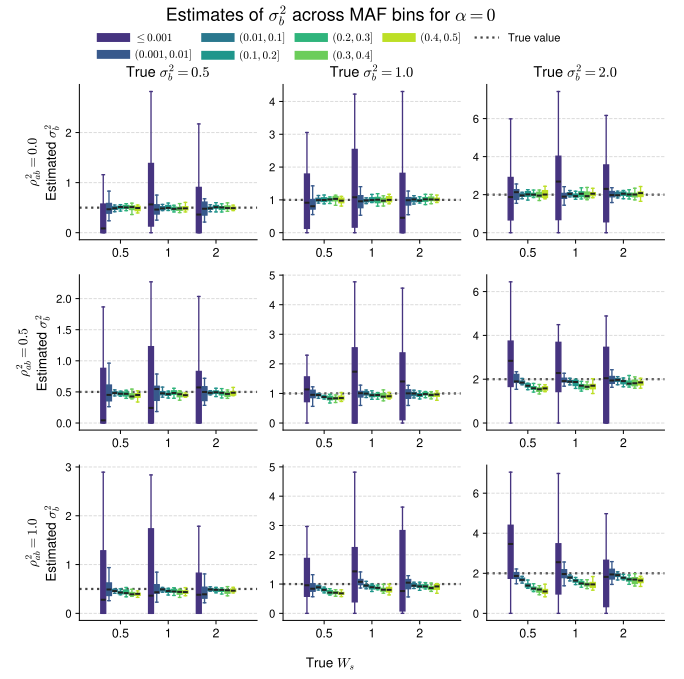


Figure A7 Bin-specific variance parameters for $\alpha = 0$.

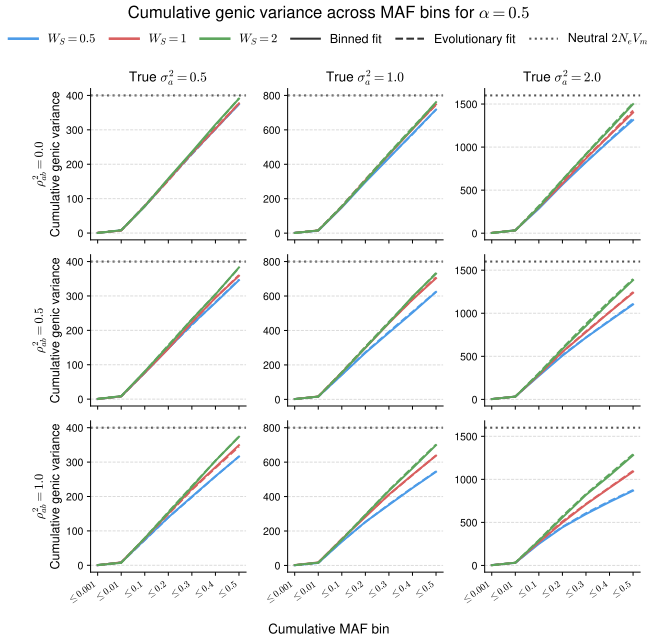


Figure A6 Cumulative genetic variance across MAF bins computed from $\alpha = 0.5$ -model with MAF stratification.

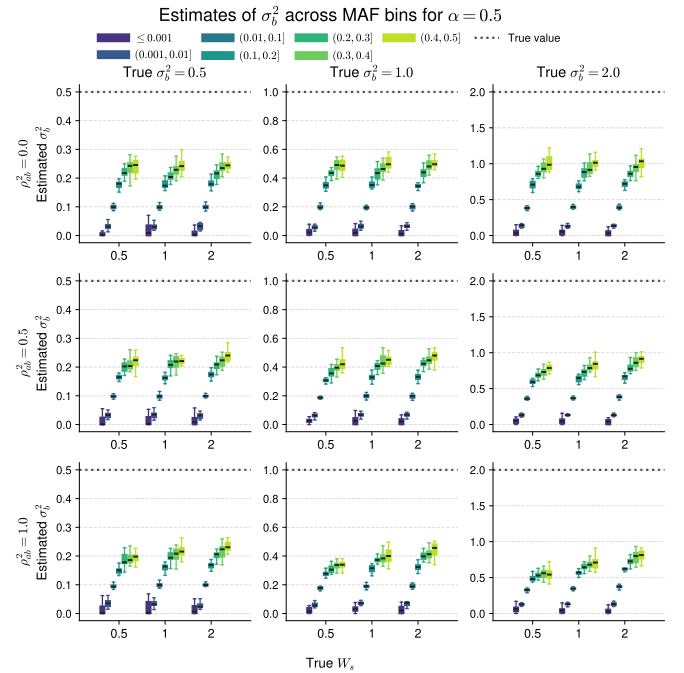


Figure A8 Bin-specific variance parameters for $\alpha = 0.5$.

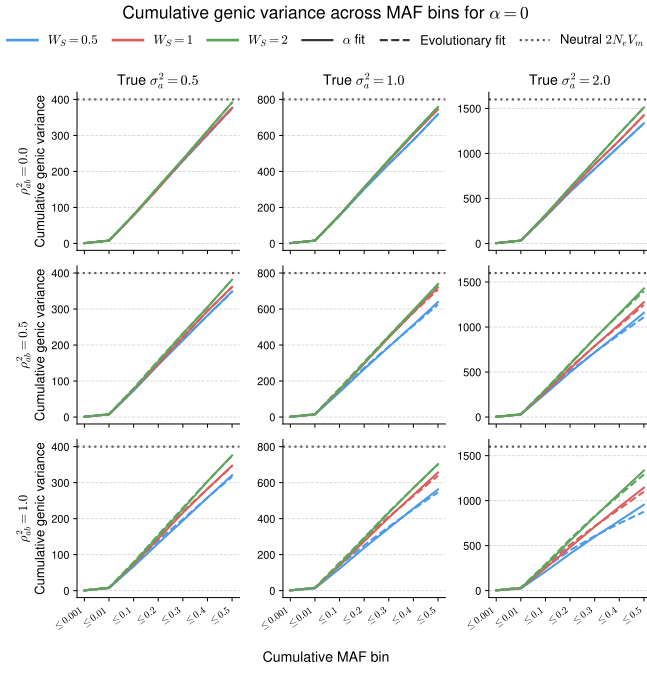


Figure A9 Cumulative genetic variance across MAF bins computed from $\alpha = 0$ -model without MAF stratification.

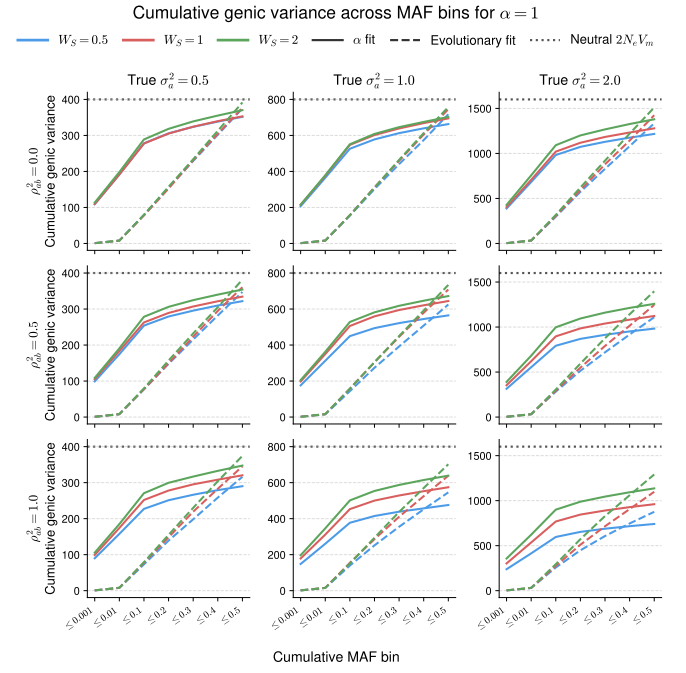


Figure A11 Cumulative genetic variance across MAF bins computed from $\alpha = 1$ -model without MAF stratification.

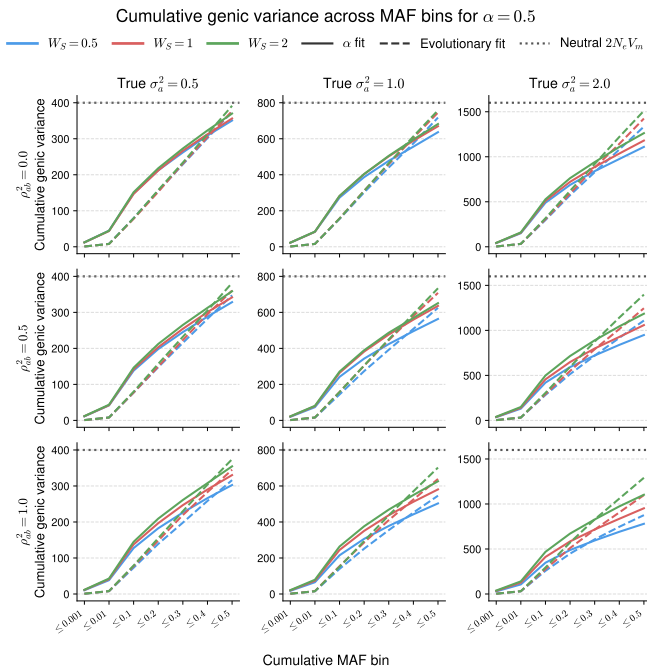


Figure A10 Cumulative genetic variance across MAF bins computed from $\alpha = 0.5$ -model without MAF stratification.